SYNTHETIC DEMO – NOT RESULTS

A. Forecast RMSE (pp) on held-out conditions, mean ± SD over 5 seeds

Arm	Test B (n=40)	Test C (n=13)	S2 (n=19)	ISU (n=63)	SNL (n=23)
A1 PINN-KAN	1.07 ± 0.01	1.27 ± 0.02	**0.70 ± 0.01**	**1.17 ± 0.05**	**1.32 ± 0.05**
A2 Hard-ODE KAN	1.19 ± 0.01	–	–	1.32 ± 0.05	1.38 ± 0.06
A3 Parametric	1.36 ± 0.01	–	0.99 ± 0.01	–	–
A4 Additive GAM	1.19 ± 0.02	–	–	1.45 ± 0.06	1.47 ± 0.06
A5 Interaction GAM	1.06 ± 0.01	–	–	1.40 ± 0.06	1.52 ± 0.06
A7 MLP stress map	1.30 ± 0.01	–	–	1.54 ± 0.06	1.64 ± 0.07
A9 Data-only F	1.93 ± 0.03	2.46 ± 0.05	1.14 ± 0.03	2.24 ± 0.09	2.12 ± 0.08
A10 Semi-empirical	1.44 ± 0.02	1.67 ± 0.04	0.84 ± 0.03	1.85 ± 0.07	1.82 ± 0.07
A11 GP-ARD	1.10 ± 0.01	1.41 ± 0.06	–	1.28 ± 0.05	1.50 ± 0.06
A12 LightGBM	1.12 ± 0.01	1.38 ± 0.03	–	1.40 ± 0.06	1.67 ± 0.07
A13 Cycling only	**1.02 ± 0.01**	–	–	–	–
A14 Storage only	–	–	0.84 ± 0.01	–	–
R1 Persistence	3.07 ± 0.00	2.72 ± 0.00	1.87 ± 0.00	5.22 ± 0.21	5.38 ± 0.22
R2 Window-linear	1.33 ± 0.00	**1.06 ± 0.00**	1.32 ± 0.00	3.95 ± 0.16	3.96 ± 0.16

B. Comparison with PINN-KAN, cost and reference ratios

	Test B		Test C		Fit time	Parameters	Test B E/R	
Arm	Ratio [95% int.]	Tier	Ratio [95% int.]	Tier	(s)	(count)	LOO	WARP
A1 PINN-KAN	–		–		7.1	2621	1.36	1.33
A2 Hard-ODE KAN	1.13 [0.99, 1.30]	desc.	–		1.5	188	1.51	1.48
A3 Parametric	1.30 [1.06, 1.52]	desc.	–		4.2	410	1.73	1.69
A4 Additive GAM	1.13 [0.95, 1.31]	desc.	–		5.5	2623	1.50	1.47
A5 Interaction GAM	1.15 [0.97, 1.24]	N	–		6.3	3312	1.35	1.32
A7 MLP stress map	1.22 [0.99, 1.37]	desc.	–		5.9	2790	1.65	1.61
A9 Data-only F	1.84 [1.31, 2.27]	S	2.28 [0.86, 2.99]	–	9.5	50945	2.45	2.39
A10 Semi-empirical	1.47 [1.08, 1.76]	S	1.09 [0.60, 2.03]	–	2.3	11	1.83	1.79
A11 GP-ARD	1.02 [0.92, 1.22]	n.t.	0.80 [0.36, 1.55]	–	3.1	14	1.40	1.37
A12 LightGBM	1.08 [0.86, 1.32]	n.t.	1.05 [0.62, 1.45]	–	0.7	1850	1.42	1.39
A13 Cycling only	0.97 [0.82, 1.07]	desc.	–		6.6	2621	1.29	1.26
A14 Storage only	–		–		–	–	–	–
R1 Persistence	3.96 [2.59, 4.77]	ref.	1.93 [1.79, 3.12]	S	0.0	0	3.89	3.81
R2 Window-linear	1.74 [1.25, 2.04]	ref.	0.79 [0.46, 1.37]	–	0.0	2	1.69	1.65

Table 1: SYNTHETIC DEMO – NOT RESULTS. Held-out accuracy of all arms. Panel A: forecast RMSE in percentage points (pp) averaged over held-out conditions (the unit); ± is the SD of that mean over the 5 final-refit seeds (reproducibility, not a confidence interval); best value in bold, second best underlined. ISU and SNL are secondary and not blind. Panel B: median over conditions of the error ratio comparator/PINN-KAN (above 1: PINN-KAN better) with its exact 95 % order-statistic interval; tier from the preregistered sign test (S superior, N non-inferior, – neither; n.t. gated and not tested; desc. descriptive); E/R is the Test B error divided by the replicate spread (LOO) or the window-aware replicate predictor (WARP), descriptive only.